

## SIMATS ENGINEERING

### ASSESSMENT TOOL 3

**COURSE NAME:** Database Management Systems  
**COURSE CODE:** CSA05

#### COURSE OUTCOME COVERED

**CO1:** *Analyze DBMS architecture, different data models, schema types, and design ER/EER diagrams, relational schemas for case-based scenarios by identifying entities and relationships. (BL1, BL2)*

Activity Tool : Technical Presentation (Low)  
Assessment Tool Weightage: 35%

Dr

| QUESTIONS |                                                                                                                         | Total Marks: 50  |       |
|-----------|-------------------------------------------------------------------------------------------------------------------------|------------------|-------|
| Q.No      | Question                                                                                                                | Bloom's Level    | Marks |
| 1         | Prepare a 5-slide presentation introducing DBMS: definition, objectives, advantages, and real-world applications.       | BL2 – Understand | 5     |
| 2         | Present a comparison between file-based systems and DBMS using real-life case studies from banking or healthcare.       | BL2 – Understand | 5     |
| 3         | Deliver a presentation explaining the three-level ANSI/SPARC architecture using diagrams and practical examples.        | BL2 – Understand | 5     |
| 4         | Present the different data models (Hierarchical, Network, Relational) with diagrams and industry use cases.             | BL1 – Remember   | 5     |
| 5         | Create and present an ER diagram for an Online Library System, explaining each component clearly.                       | BL2 – Understand | 5     |
| 6         | Prepare a presentation on the EER model, demonstrating generalization and specialization with a University case study.  | BL2 – Understand | 5     |
| 7         | Present the concept of data independence (logical and physical) and its significance in DBMS architecture.              | BL2 – Understand | 5     |
| 8         | Deliver a technical presentation on the software components of a DBMS, explaining query processor, storage manager, and | BL1 – Remember   | 5     |

|    |                                                                                                    |                  |   |
|----|----------------------------------------------------------------------------------------------------|------------------|---|
|    | transaction manager.                                                                               |                  |   |
| 9  | Prepare a presentation comparing strong and weak entities in ER modeling with real-world examples. | BL2 – Understand | 5 |
| 10 | Present a complete design walkthrough of an EER diagram for a Retail Chain Management System.      | BL2 – Understand | 5 |

**Code of Conduct:**

**R.NIHARIKA(192511221)**

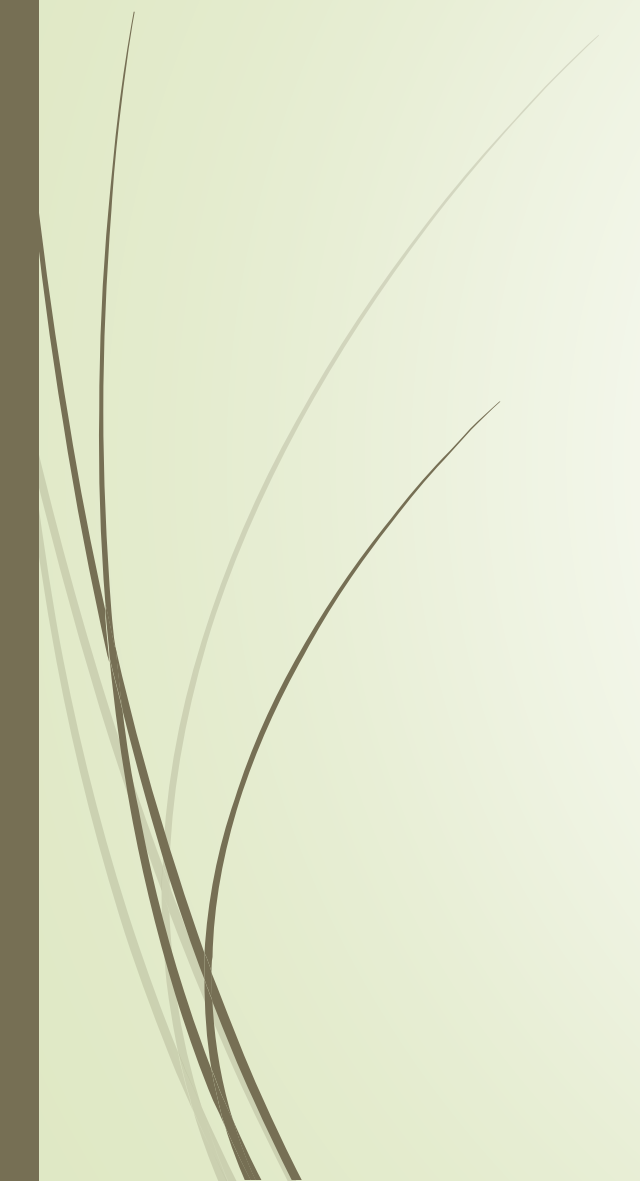
I \_\_\_\_\_ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

**Signature of the student:**

*R. Niharika*

**RUBRICS FOR CALCULATION:**

| Technical Presentation     |           |                                                                    |                                                                |                                                            |                                                                     |
|----------------------------|-----------|--------------------------------------------------------------------|----------------------------------------------------------------|------------------------------------------------------------|---------------------------------------------------------------------|
| Criteria                   | Weightage | Level 4 – Exemplary                                                | Level 3 – Proficient                                           | Level 2 – Developing                                       | Level 1 – Beginning                                                 |
| Technical Content Accuracy | 30        | Highly accurate, in-depth, and insightful technical explanation    | Technically correct content with adequate depth and clarity    | Basic concepts presented with limited accuracy and depth   | Content contains major technical errors; concepts poorly understood |
| Problem Identification     | 25        | Problem rigorously analyzed using appropriate and advanced methods | Problem clearly defined with a logical engineering approach    | Problem identified but analysis or methodology is weak     | Problem statement unclear or incorrect; no logical approach         |
| Use of Tools               | 20        | Advanced tools, well-analyzed data, and highly effective visuals   | Appropriate tools and data used effectively with clear visuals | Limited use of tools and basic data interpretation         | Minimal or incorrect use of tools and data; poor visuals            |
| Communication              | 15        | Highly engaging, confident, and audience-focused delivery          | Clear, organized, and professional communication               | Understandable but lacks clarity, structure, or confidence | Poor organization, unclear speech, ineffective slides               |



|                 |    |                                                              |                                                        |                                                  |                                                                  |
|-----------------|----|--------------------------------------------------------------|--------------------------------------------------------|--------------------------------------------------|------------------------------------------------------------------|
| Professionalism | 10 | Exemplary professionalism, leadership, and ethical awareness | Professional conduct with effective team collaboration | Basic professionalism; uneven team participation | Lacks professionalism; minimal contribution or ethical awareness |
|-----------------|----|--------------------------------------------------------------|--------------------------------------------------------|--------------------------------------------------|------------------------------------------------------------------|



**SIMATS**  
**ENGINEERING**



**SIMATS**  
Saveetha Institute of Medical And Technical Sciences  
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

**CSA0501-DATABASE MANAGEMENT SYSTEM**

# **TECHNICAL PRESENTATION**

**GUIDED BY**

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**PRESENTED BY**

**R.NIHARIKA(192511221)**



# 1. Introduction to DBMS



## Database Management System (DBMS)

- DBMS is software used to **store, organize, manage, and retrieve data**.
- It acts as an interface between the **user and the database**.
- It makes data easy to access, update, and manage.
- Examples: MySQL, Oracle, PostgreSQL, Microsoft SQL Server.



# Definition of DBMS



## What is DBMS?

A DBMS is a software system that allows users to create, store, retrieve, update, and manage data in a database.

### Main functions:

- Create and manage databases
- Insert and update data
- Search and retrieve information
- Delete unwanted data
- Control access to data



# Objectives of DBMS



- **Data Storage:** Store large amounts of data efficiently.
- **Data Security:** Protect data from unauthorized access.
- **Data Integrity:** Maintain accurate and consistent data.
- **Data Sharing:** Allow multiple users to access data.
- **Data Recovery:** Recover data after failures.
- **Reduce Redundancy:** Avoid unnecessary duplication of data.



# Advantages of DBMS



- Reduces **data duplication**
- Provides better **data security**
- Improves **data consistency**
- Allows **easy data access**
- Supports **multiple users**
- Provides **backup and recovery**
- Makes data **easy to maintain**
- Helps in **faster decision-making**



# Real-World Applications



| Application                                                                                  | Use of DBMS                                   |
|----------------------------------------------------------------------------------------------|-----------------------------------------------|
|  Banking    | Customer accounts, transactions & loans       |
|  Hospitals  | Customer accounts, transactions & loans       |
|  Education  | Student records, attendance & marks           |
|  Airlines | Flights, passenger records & ticket bookings  |
|  Hotels   | Room reservations, customer details & billing |



2.

## File-Based System vs DBMS



| File-Based System             | DBMS                      |
|-------------------------------|---------------------------|
| Data stored in separate files | Data stored in a database |
| More data duplication         | Less data duplication     |
| Limited security              | Better security           |
| Difficult data sharing        | Easy data sharing         |
| Difficult backup and recovery | Easy backup and recovery  |



# Real-World Case Study: Banking



## Banking Example

### File-Based System:

- Customer details stored in separate files.
- Same information may be repeated.
- Updating data is difficult.
- Data sharing is limited.

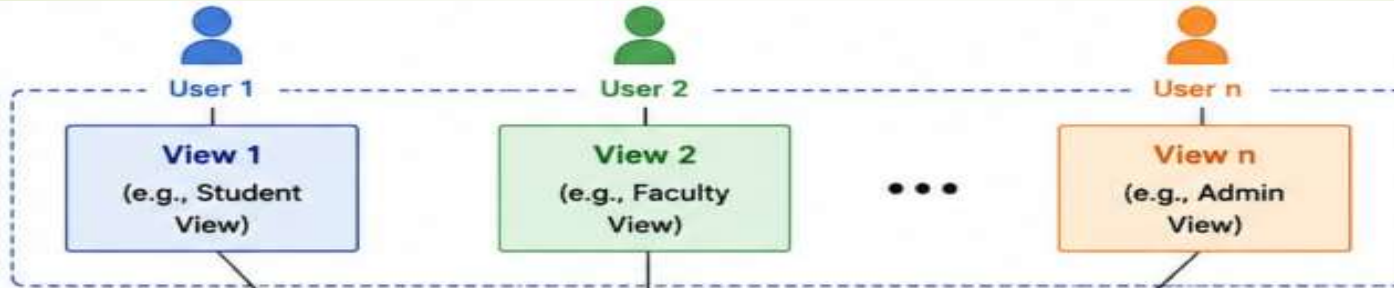
### DBMS:

- Customer, account, and transaction data are connected.
- Data can be updated easily.
- Provides better security.
- Multiple users can access data.

# 3. THREE-LEVEL ANSI/SPARC ARCHITECTURE

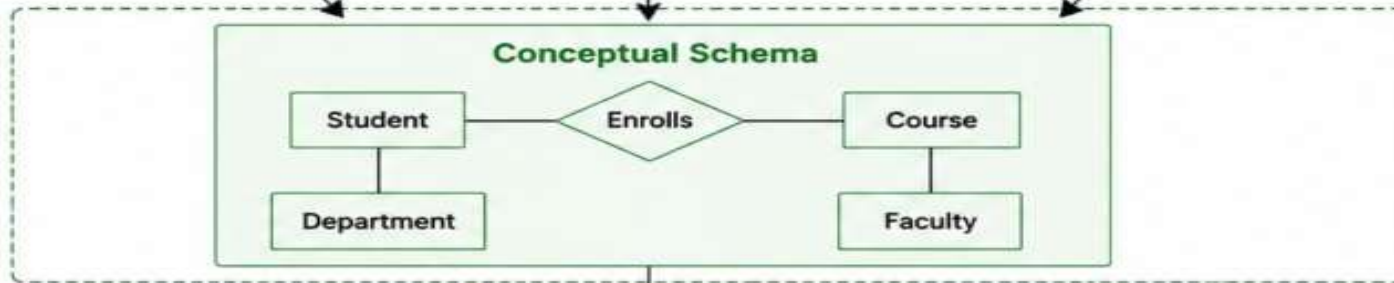
## 1. EXTERNAL LEVEL

Multiple user views of the database.



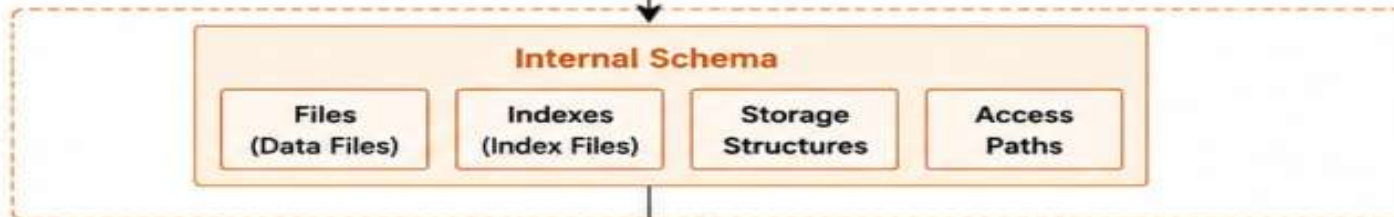
## 2. CONCEPTUAL LEVEL

Community view of the entire database. Describes logical structure and relationships.



## 3. INTERNAL LEVEL

Physical storage structure and access methods.



## KEY POINTS

- Provides data abstraction.
- Supports data independence.
- Different users can have different views.
- Changes in one level do not affect other levels.

## EXAMPLE (UNIVERSITY DBMS)

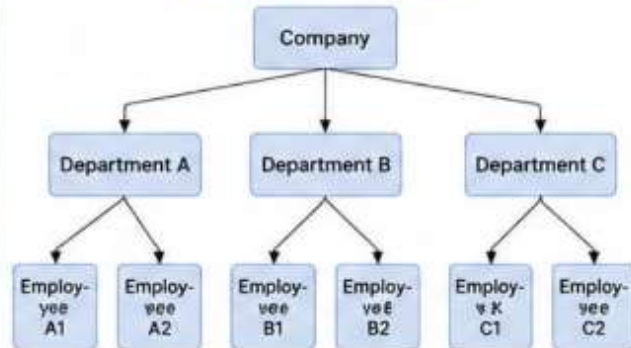
- **External Level:** Student sees marks and attendance.
- **Conceptual Level:** Database contains Student, Course, Faculty, Department and relationships.
- **Internal Level:** Data is stored in files, indexes on disk.

# 4.data models in DBMS

## 1. HIERARCHICAL MODEL

- Data is organized in a tree-like structure.
- One parent can have many children.
- Each child has only one parent.

DIAGRAM (Tree Structure)



### INDUSTRY USE CASES



**Banking:**  
Used for storing information in banking systems like account hierarchies, branches, customers.

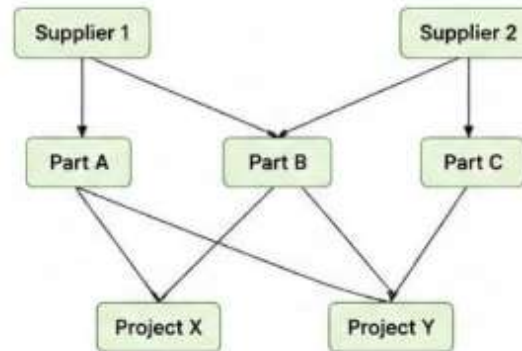


**File Systems:**  
Operating systems use hierarchical structure to organize files and directories.

## 2. NETWORK MODEL

- Data is organized as a graph structure.
- A record can have multiple parent records.
- Many-to-many relationships are supported.

DIAGRAM (Graph Structure)



### INDUSTRY USE CASES



**Airlines Reservation:**  
A flight can have many passengers and a passenger can book many flights.

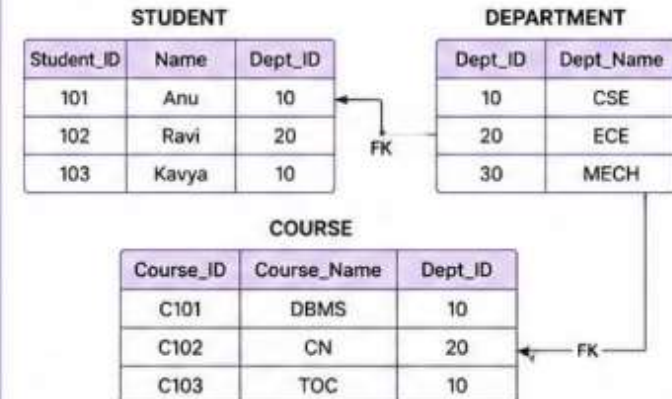


**Telecommunication:**  
Used to manage complex many-to-many relationships in networks.

## 3. RELATIONAL MODEL

- Data is organized in tables (rows and columns).
- Based on the concept of relations.
- Relationships are established using keys.

DIAGRAM (Table Structure)



### INDUSTRY USE CASES



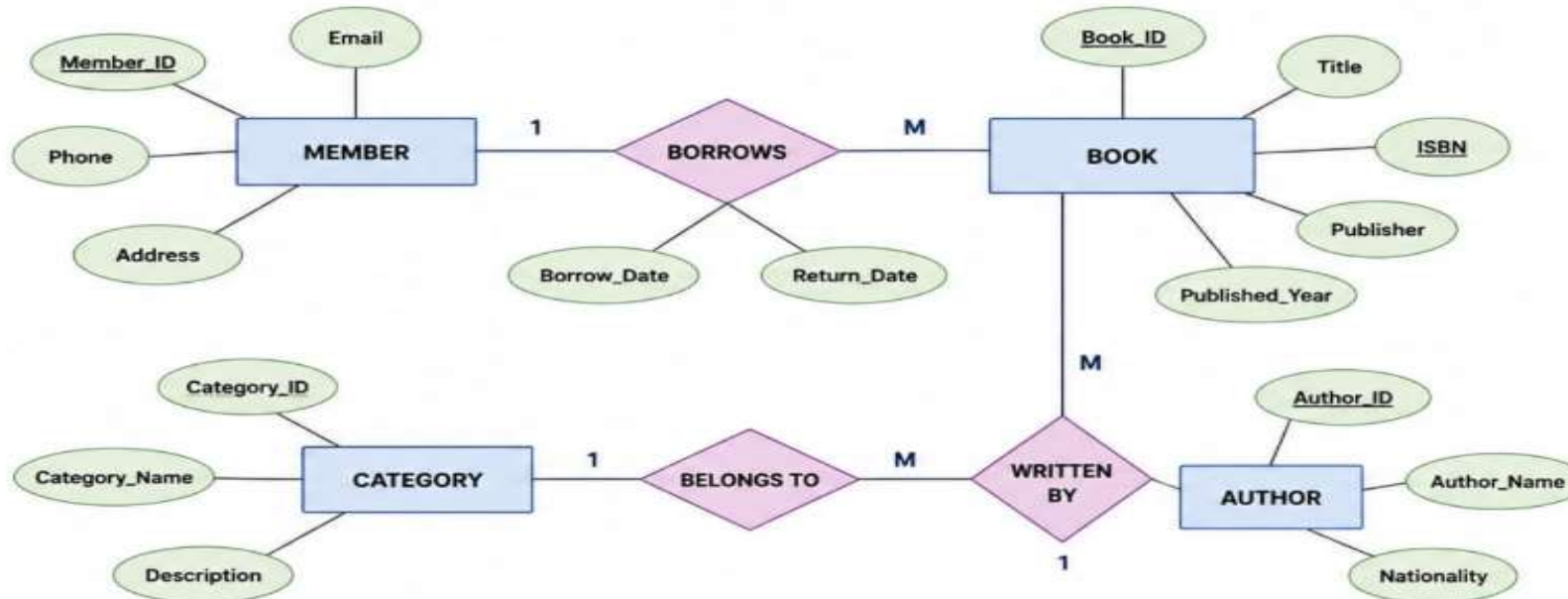
**E-Commerce:**  
Used in product catalogs, customer orders, payments, inventory, etc.



**Healthcare:**  
Stores patient details, appointments, treatments, billing in relational tables.

# 5. ER DIAGRAM FOR ONLINE LIBRARY SYSTEM

An ER (Entity–Relationship) diagram represents the data of the system in the form of entities, attributes and relationships. Online Library System stores information about books, members, authors, categories, and borrowing.



## CARDINALITY

1 = One  
M = Many  
1 — M = One-to-Many Relationship

## KEY POINTS

- The ER diagram provides a high-level view of the data and relationships in the system.
- It helps in designing the database structure before implementation.
- This model ensures data organization, reduces redundancy, and maintains relationships.

## USE CASE



This ER model is used in online library systems to manage members, books, authors, categories, and borrowing operations efficiently.

## COMPONENTS EXPLAINED



### ENTITY

A real-world object or concept about which data is stored.



### ATTRIBUTE

Properties or characteristics of an entity.



### RELATIONSHIP

Association between two or more entities.

### KEY ATTRIBUTE (Underlined)

Uniquely identifies each instance of an entity.

## ENTITIES IN THE DIAGRAM

- MEMBER** – Library members who borrow books.
- BOOK** – Books available in the library.
- AUTHOR** – Authors who write books.
- CATEGORY** – Category or genre of books.

## RELATIONSHIPS IN THE DIAGRAM

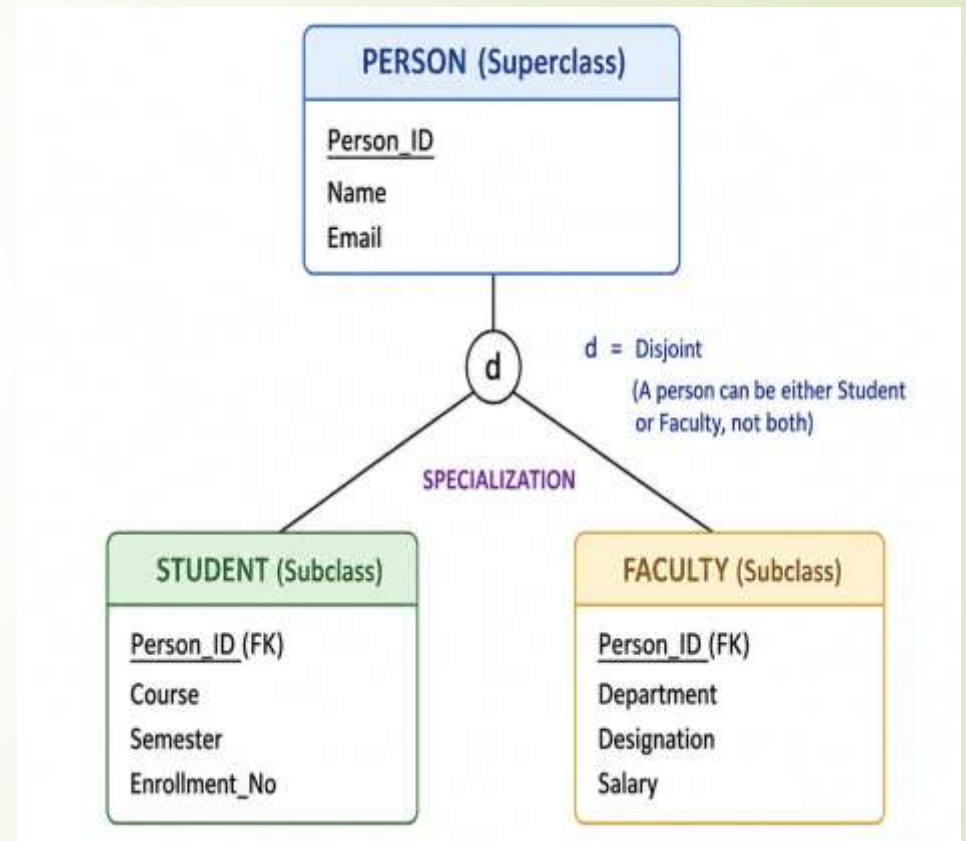
- MEMBER BORROWS BOOK** – Members can borrow many books; each book can be borrowed by many members over time.
- BOOK WRITTEN BY AUTHOR** – An author writes many books; a book has one author.
- BOOK BELONGS TO CATEGORY** – A category has many books; a book belongs to one category.

## 6. EER Model – University Case Study

- **EER (Enhanced Entity-Relationship) Model** extends the basic ER model.
- **Generalization:** Combines similar entities into a common superclass.
- **Specialization:** Divides a superclass into specific subclasses.

### University Example:

- **Person** → Superclass
- **Student** and **Faculty** → Specialized subclasses
- Common attributes: **Person\_ID, Name, Email**
- Student-specific: **Course, Semester**
- Faculty-specific: **Department, Salary**

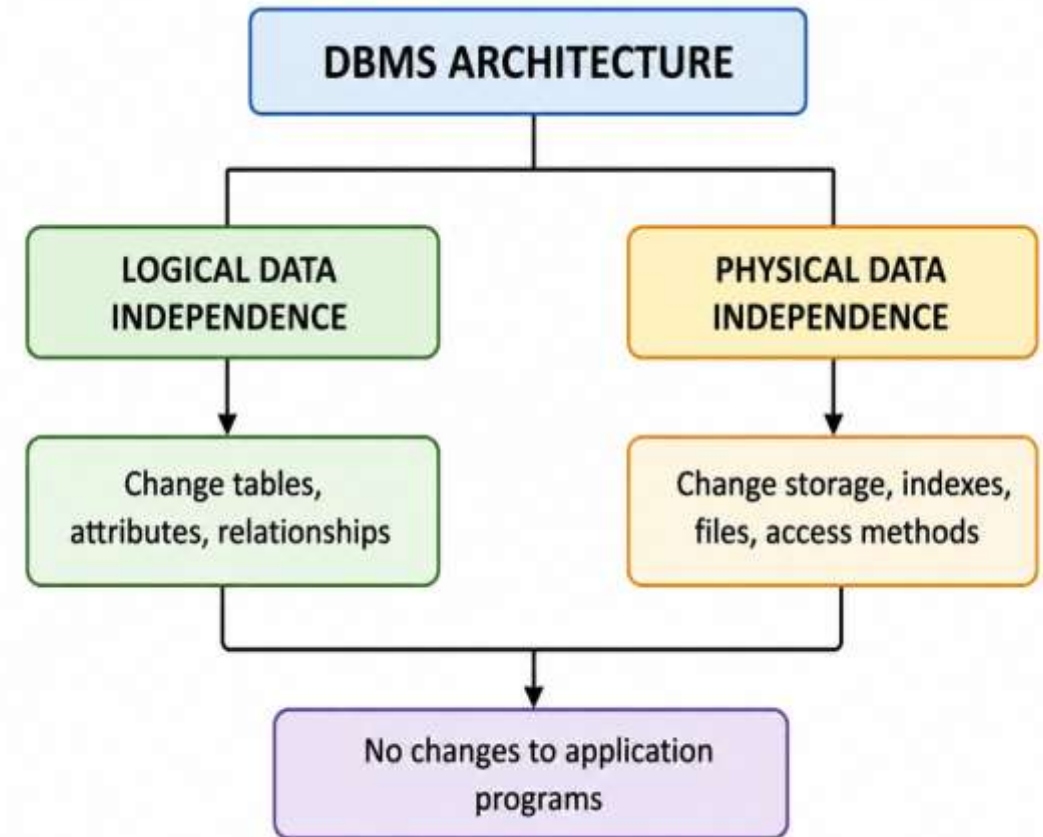


## 7. Data Independence in DBMS

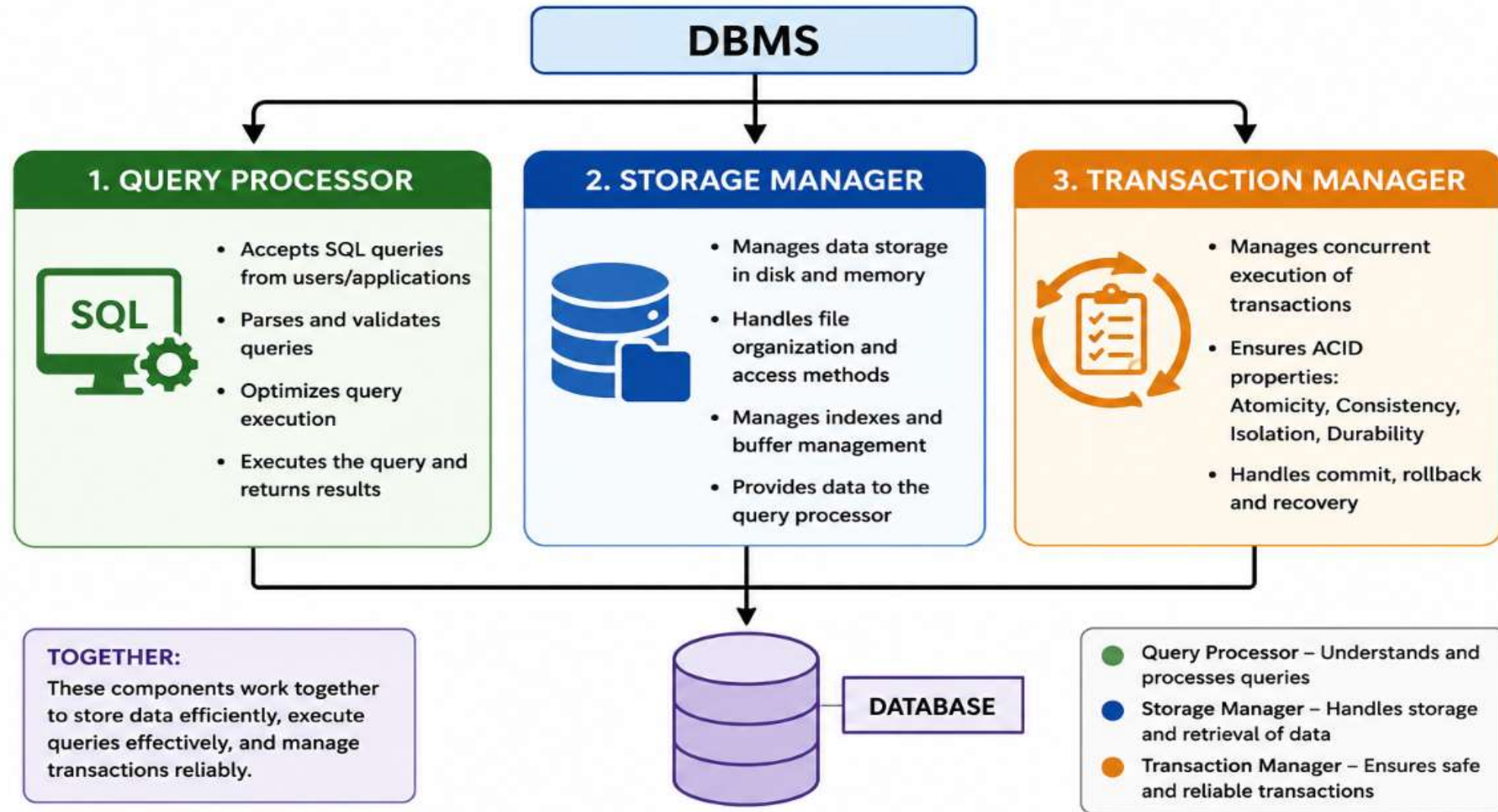
**Data Independence:** Ability to change the database structure at one level without affecting the next higher level.

**Significance:**

- Reduces maintenance effort
- Improves flexibility
- Protects applications from database changes
- Makes DBMS easier to manage and upgrade



# 8. Software Components of DBMS





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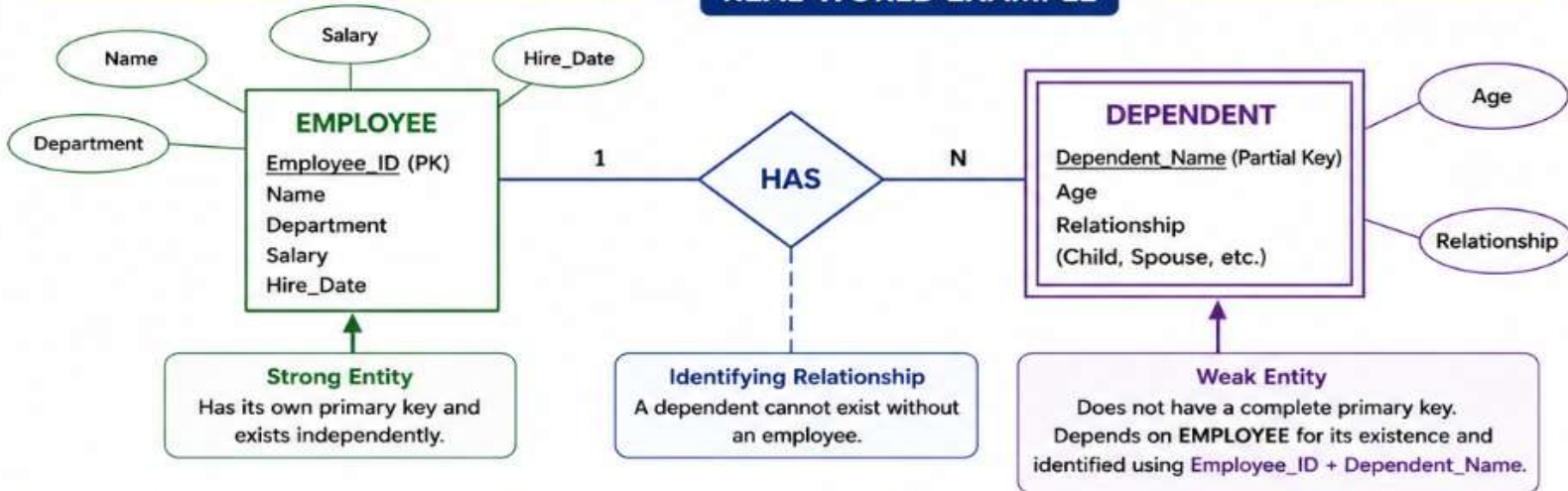
# 9. Strong vs Weak Entities in ER Model



Comparison with Real-World Example

| STRONG ENTITY                                     |                | WEAK ENTITY                                                                                                         |  |
|---------------------------------------------------|----------------|---------------------------------------------------------------------------------------------------------------------|--|
| ✓ Has its own primary key                         | Key            | ✗ Does not have a complete primary key                                                                              |  |
| ✓ Exists independently                            | Existence      | ✗ Depends on a strong entity for its existence                                                                      |  |
| ✓ Represented by single rectangle                 | Notation       | Represented by double rectangle  |  |
| ✓ Can have relationships with other entities      | Relationship   | Has an identifying relationship with owner entity                                                                   |  |
| ✓ Example: <b>EMPLOYEE</b>                        | Example        | Example: <b>DEPENDENT</b>                                                                                           |  |
| ✓ Identified by its own primary key (Employee_ID) | Identification | Identified by combination of owner key + partial key (Employee_ID + Dependent_Name)                                 |  |

REAL-WORLD EXAMPLE



## KEY POINTS

- **EMPLOYEE** is a strong entity because it has its own primary key (Employee\_ID).
- **DEPENDENT** is a weak entity because it depends on **EMPLOYEE** and is identified by (Employee\_ID + Dependent\_Name).
- If an employee is removed, all his/her dependents cannot exist.

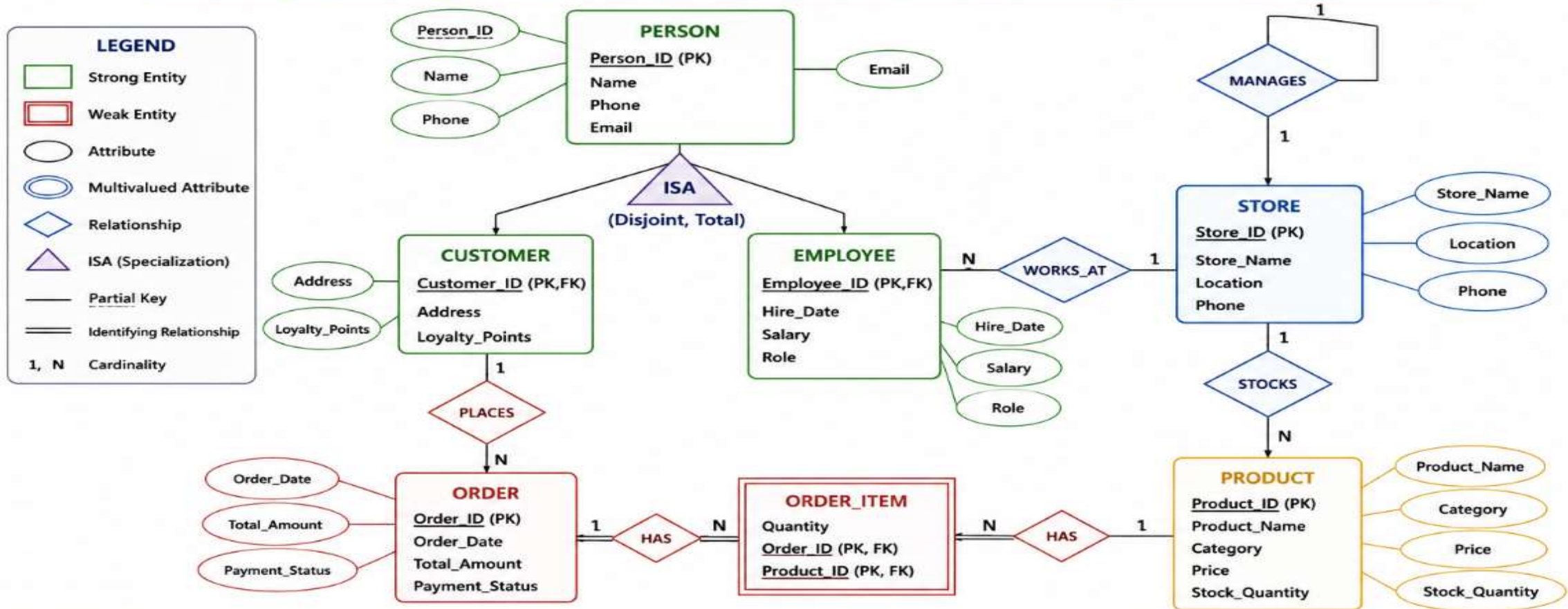
-  Strong Entity
-  Weak Entity
-  Relationship
-  Attribute
- 1, N Cardinality



Real-World Example: In a company, an **EMPLOYEE** can have one or more **DEPENDENT**s (e.g., children, spouse). A **DEPENDENT** cannot exist without the **EMPLOYEE**.

# 10. EER Diagram – Retail Chain Management System

**Design Goal:** Manage customers, employees, stores, products, orders and inventory across a retail chain.



## DESIGN WALKTHROUGH

- PERSON is a supertype with two subtypes CUSTOMER and EMPLOYEE (Disjoint, Total).
- EMPLOYEE works at a STORE. Each store can have many employees. A store is managed by one employee.
- STORE stocks many PRODUCTS. Each product is stocked in one store.
- CUSTOMER places ORDERS. Each order has many ORDER\_ITEMS.
- PRODUCT appears in many ORDER\_ITEMS. ORDER\_ITEM is a weak entity identified by ORDER and PRODUCT.

## KEY CONSTRAINTS

- Order cannot exist without a Customer.
- Order\_Item cannot exist without an Order and a Product.
- Employee must work at one Store.
- Each Store must be managed by one Employee.